Streamlining IT Procurement: Automating Standard Laptop Orders with WI

A Final Year Project Document

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1. Abstract

This project focuses on simplifying the way standard laptop orders are handled inside an organisation. At present, most companies still depend on emails, paper forms and manual checks when an employee needs a new laptop. This process is slow, error-prone and hard to track. The proposed system uses a simple web-based workflow (called WI – Workflow Interface) that automatically guides a laptop request from the moment it is raised until the device is delivered. Key steps such as manager approval, budget verification, vendor selection and inventory update are performed by the system itself. The result is faster delivery, clearer records and less work for the IT and purchase teams. The solution is designed for final-year students to understand how automation can improve everyday IT procurement tasks.

2. Introduction

Every modern company needs computers for its staff. When a new employee joins or an old laptop becomes slow, the IT department must order a replacement. In many places this still happens through long email chains and printed approval forms. The request may sit for days waiting for a manager's signature or for the finance team to confirm that money is available. Sometimes the same laptop model is ordered twice because nobody checked the stock list. These delays waste time and money.

This project suggests a better way. We design a simple automated workflow that takes care of the whole laptop-ordering cycle. The system is named WI (Workflow Interface). Once a user raises a request, WI moves the request through each required stage automatically and keeps everyone informed. The goal is to make standard laptop procurement fast, transparent and easy to audit.

3. Problem Statement

The current laptop procurement process in most organisations suffers from several practical problems:

- Requests are raised by email or paper form, so there is no single place to track status.
- Manager approval is often delayed because the approver does not receive a clear reminder.
- Budget checks are done manually; sometimes orders are placed when funds are already exhausted.
- Vendor selection and price comparison take extra days of back-and-forth communication.
- Inventory is not updated in real time, leading to duplicate orders or stock-outs.
- There is no automatic record of who approved what and when, making audits difficult.

Because of these issues, a standard laptop order can take two to three weeks instead of a few days. The project aims to remove these bottlenecks by introducing an automated workflow.

4. Objectives

The main objectives of this project are:

1. To design a simple web-based interface (WI) where any employee can raise a standard laptop request.
2. To implement automatic routing of the request to the correct manager for approval.
3. To add a real-time budget check so that orders are blocked if funds are insufficient.
4. To maintain a small catalogue of approved laptop models and preferred vendors.
5. To update inventory automatically when a laptop is ordered or delivered.
6. To keep a complete digital trail of every step for future audits and reports.
7. To reduce the average time taken for a standard laptop order from weeks to a few working days.

5. Existing System

In the existing system the process is almost completely manual. An employee writes an email or fills a printed form describing the need for a laptop. The form is then sent to the reporting manager. After the manager signs it, the paper or email moves to the IT procurement team. The procurement officer checks the budget with the finance department by phone or another email. Once the budget is confirmed, the officer contacts two or three vendors, waits for quotations, selects one vendor and places the order. When the laptop arrives, someone manually updates an Excel sheet that acts as the inventory list. If any person is on leave, the whole chain stops. There is no dashboard to see where a request is stuck. This system works for very small companies but becomes slow and confusing as the number of employees grows.

6. Proposed System

The proposed system replaces the manual chain with a single automated workflow called WI. All standard laptop requests start and end inside this system. The main features are:

• A simple online form where the user selects the laptop type (for example: standard office laptop, developer laptop) and enters basic details.

• Automatic assignment of the request to the correct reporting manager based on the employee's department.

• Email and dashboard notifications so that the manager can approve or reject with one click.

• Instant budget verification against the department's remaining laptop budget.

• A short list of pre-approved vendors and models so that price comparison is done by the system itself.

• Automatic generation of a purchase order once all checks pass.

• Real-time update of the inventory database when the order is placed and when the device is received.

• A complete history log that shows every action taken on the request.

Because the rules are fixed for standard models, most requests can move from submission to order placement without any human delay beyond the manager's single approval click.

7. System Architecture & Workflow

The system is built as a simple three-layer architecture:

1. **Presentation Layer** – A web form and a small dashboard that employees and managers use.

2. **Business Logic Layer** – The WI engine that applies the rules for approval, budget and vendor selection.
3. **Data Layer** – A database that stores requests, user details, budget figures, vendor list and inventory.

Step-by-step workflow for a standard laptop order:

Step 1 – Raise Request

Employee logs into WI, chooses “Standard Laptop”, selects model from the approved list, enters reason and submits. A unique request ID is generated and the status becomes “Pending Manager Approval”.

Step 2 – Manager Approval

The system sends an email and shows the request on the manager’s dashboard. The manager can Approve, Reject or Ask for more information. If approved, the request moves to the next stage automatically.

Step 3 – Budget Check

WI reads the remaining budget for that department. If the cost of the selected laptop is within the budget, the amount is reserved and the request continues. If not, the request is returned to the employee with a clear message.

Step 4 – Vendor Selection

The system looks at the pre-approved vendor list for the chosen model, picks the vendor with the best current price and delivery time, and prepares a purchase order.

Step 5 – Order Placement

The purchase order is sent to the selected vendor by email or API. Status changes to “Order Placed”. Finance and IT teams receive a copy for their records.

Step 6 – Inventory Update

When the vendor confirms dispatch, the inventory record is updated to “In Transit”. When the laptop is received and checked by the IT team, the status becomes “In Stock / Assigned” and the employee is notified.

Step 7 – Closure

The request is marked “Completed”. All actions (who approved, when, which vendor, serial number) are stored permanently for audit.

8. Benefits

- Faster turnaround – most standard orders finish in 2–4 working days instead of 2–3 weeks.
- Less manual work – IT and purchase staff only handle exceptions, not every routine request.
- Better budget control – no order can be placed without a live budget check.
- Clear visibility – anyone can see the current status of their request on the dashboard.
- Accurate inventory – stock figures stay up to date automatically.
- Easy audit trail – every approval and decision is recorded with date and user name.
- Reduced errors – no lost emails or wrong model numbers because the catalogue is controlled.

9. Conclusion

Automating standard laptop orders through a simple workflow interface removes many of the delays and mistakes that exist in the traditional paper-and-email process. By combining manager approval, budget verification, vendor selection and inventory update into one continuous flow, the organisation can deliver laptops to employees much faster while keeping full control and clear records. The solution is practical for final-year implementation because it uses well-known web technologies and focuses on a single, well-defined

process. Once the basic system is running, it can be extended to other IT items such as monitors or mobile phones.

10. Future Scope

- Add mobile app support so managers can approve requests from their phones.
- Integrate with the company’s ERP or finance software for automatic payment tracking.
- Include non-standard or high-value devices with multi-level approval.
- Use simple machine-learning rules to suggest the best laptop model based on the user’s role.
- Generate monthly reports that show average delivery time, budget usage and popular models.
- Connect directly to vendor portals so order status is updated without manual entry.

11. References

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3. Official documentation of workflow engines and web frameworks used in academic projects.

4. Sample procurement process guidelines published by various IT service companies (public domain material).

5. Final-year project reports on automation of administrative workflows available in college libraries.